ERGA Assembly Report

v24.10.15

Tags: ERGA-Community

TxID	880251
ToLID	**fZinAsp1**
Species	Zingel asper
Class	Actinopteri
Order	Perciformes

Genome Traits	Expected	Observed
Haploid size (bp)	843,563,058	963,127,775
Haploid Number	24 (source: ancestor)	24
Ploidy	2 (source: ancestor)	2
Sample Sex	male	unknown

EBP metrics summary and curation notes

Obtained EBP quality metric for pri: 6.7.Q55

The following metrics were automatically flagged as below EBP recommended standards or different from expected:

. Observed sex is different from Sample sex

. Assembly length loss > 3% for pri

Curator notes

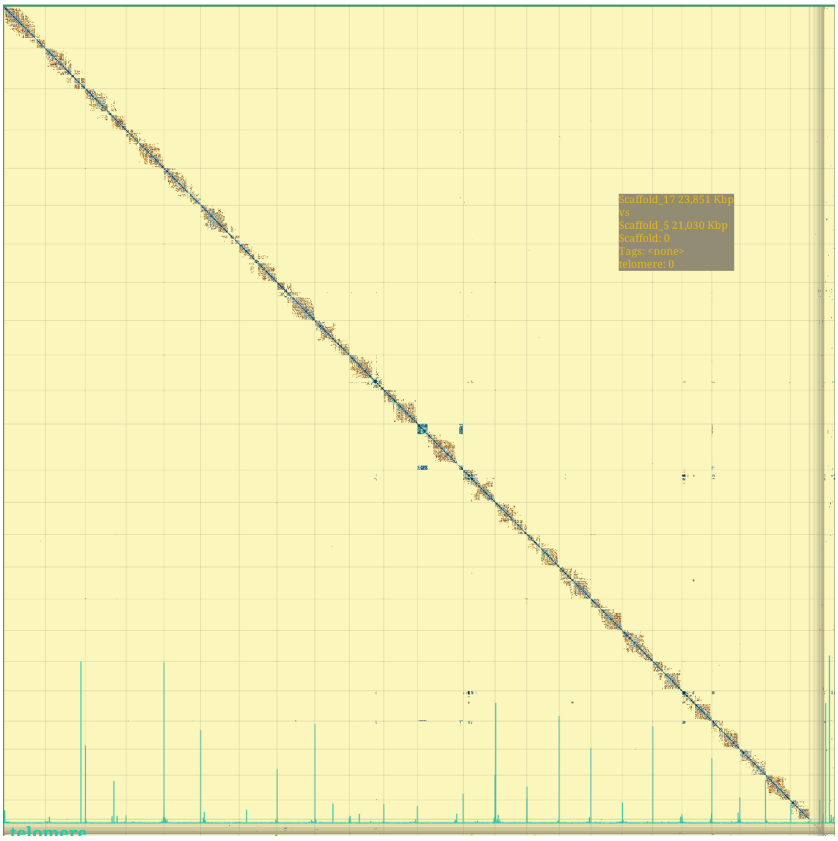
. Interventions/Gb:
. Contamination notes: "Blobtools detected presence of bacteria contaminants, which were removed. Mitochondrial genome was removed from the assembly"
. Other observations: "Sex chromosomes not found, Zingel zingel, x1x2y - x1x2, GC content scaffolds that were not between 0,38 and 0,463, were smaller than 14400bp and had less coverage than 6,4x were filtered out through blobtools. Also, those sequences whose blast results were assigned to other lineages than Chordata were spurious and therefore, removed.Manual curation using Hi-C contact maps resulted in 21 joins and rearrangement of the assembly. The assembly contained 293 scaffolds before manual curation and 245 scaffolds in the final curated assembly."

Quality metrics table

Metrics	Pre-curation pri	Curated pri
Total bp	1,014,352,583	963,127,775
GC %	41.09	40.86
Gaps/Gbp	0	848.28
Total gap bp	0	152,100
Scaffolds	1,747	245
Scaffold N50	3,849,056	39,905,744
Scaffold L50	68	11
Scaffold L90	310	22
Contigs	1,747	1,062
Contig N50	3,849,056	3,215,671
Contig L50	68	88
Contig L90	310	330
QV	55.6548	55.8789
Kmer compl.	95.7086	95.3343
BUSCO sing.	97.8%	98.4%
BUSCO dupl.	0.9%	0.7%
BUSCO frag.	0.7%	0.2%
BUSCO miss.	0.6%	0.7%

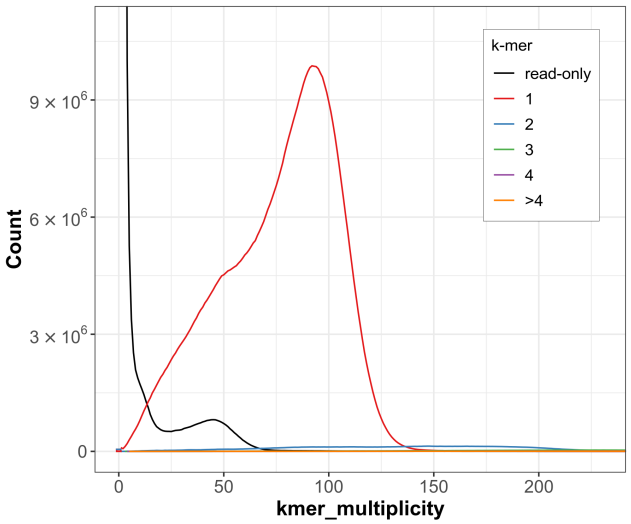
BUSCO: 6.0.0 (euk_genome_min, miniprot) / Lineage: actinopterygii_odb12 (genomes:75, BUSCOs:7207)

HiC contact map of curated assembly

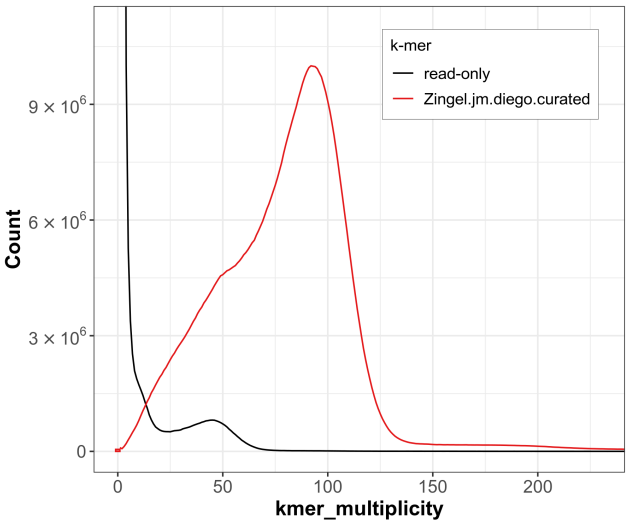


pri [LINK]

K-mer spectra of curated assembly

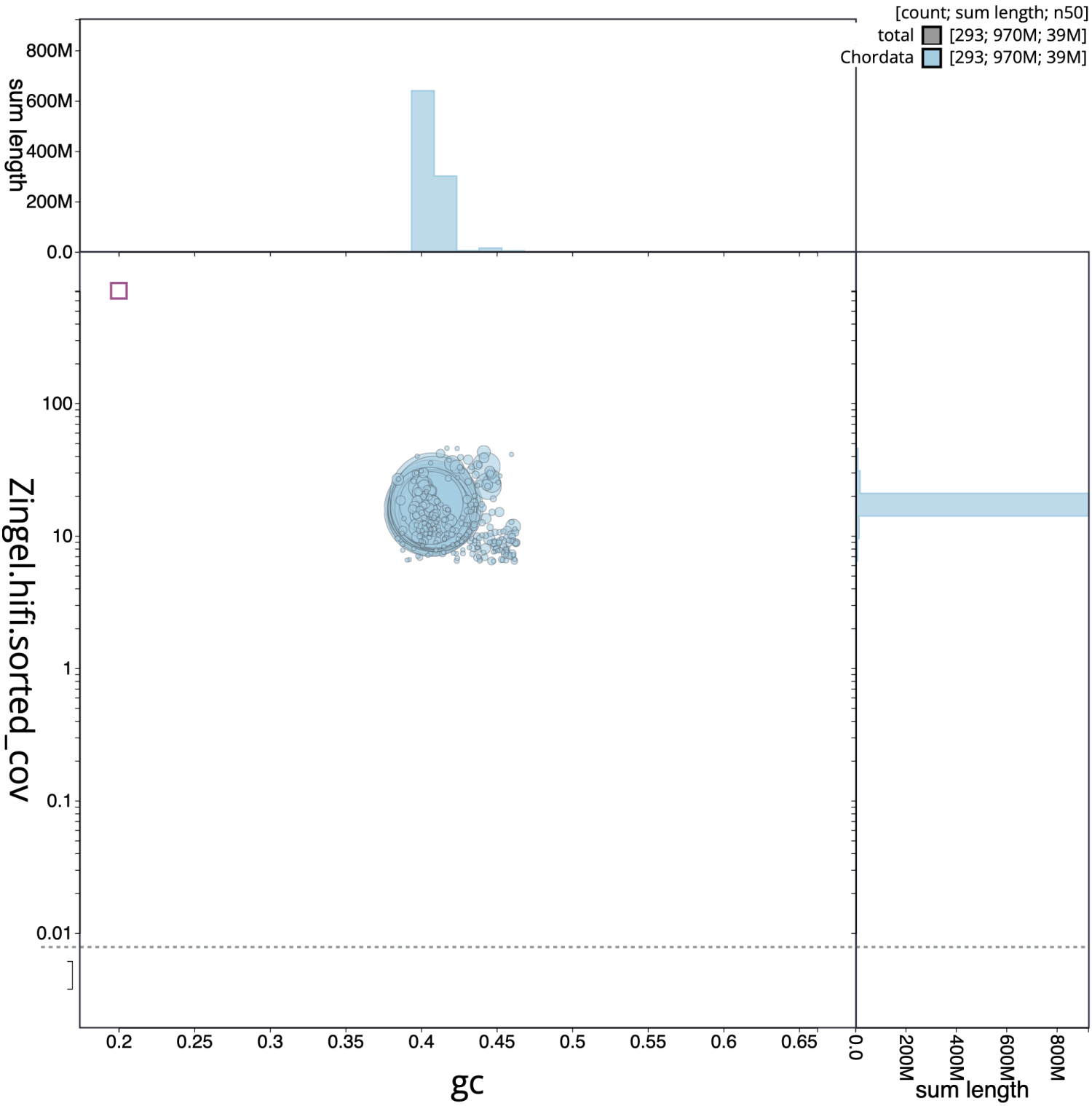


Distribution of k-mer counts per copy numbers found in asm



Distribution of k-mer counts coloured by their presence in reads/assemblies

Post-curation contamination screening



pri. Bubble plot circles are scaled by sequence length, positioned by coverage and GC proportion, and coloured by taxonomy. Histograms show total assembly length distribution on each axis.

Data profile

Data	HiFi	Bionano	HiC
Coverage	40x	NA	60x

Assembly pipeline

- **Hifiasm**
  |_ *ver:* 0.25.40
  |_ *key param:* primary
- **purge_dups**
  |_ *ver:* 1.2.5
  |_ *key param:* NA
- **YaHS**
  |_ *ver:* 1.2.2
  |_ *key param:* NA

Curation pipeline

- **BlobtoolKit**
  |_ *ver:* 2.1.1
  |_ *key param:* NA

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